

3 Ans: D

Since the ball bearing is released from rest, its initial speed is zero. Hence at $t = 0$ s, it is a stationary point on the graph.

The ball bearing accelerates downwards under the pull of gravity until it enters the oil. During this part of the motion, the gradient of the graph increases.

In the oil, the ball bearing will experience a resistive force upwards that is greater than its weight. This will cause the ball bearing to decelerate. During this part of the motion, the gradient of the graph decreases.

When the resistive force decreases (speed is decreasing) until it is equal in magnitude to its weight, the ball bearing will travel downwards with constant speed until it hits the bottom of the cylinder and comes to a sudden stop. During this part of the motion, the gradient of the graph is constant and will become zero abruptly.

4 Ans: C